



CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION
ENVIRONMENTAL SCIENCE

UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

PAPER 02

2 hours 30 minutes

04 MAY 2010 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Candidates **MUST** answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.

NOTHING HAS BEEN OMITTED

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MODULE 1

Answer ANY TWO questions.

1. (a) Distinguish between 'primary ecological succession' and 'secondary ecological succession'. [3 marks]
- (b) Figure 1 shows the population of alligators in the Black River over a twelve-year period. Study Figure 1 and answer the questions that follow.

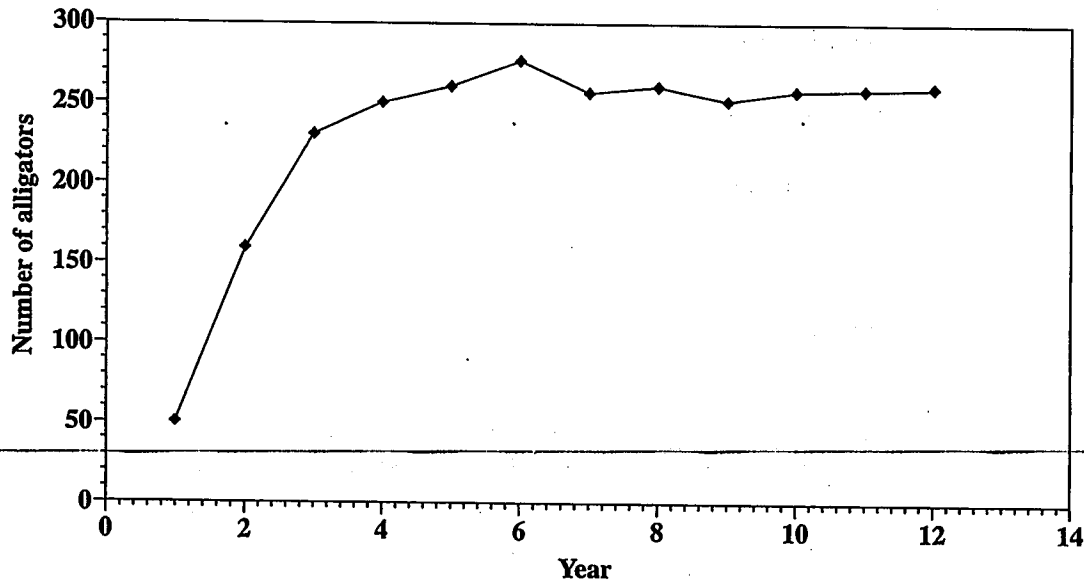


Figure 1. Population of alligators in the Black River

- (i) Describe the trend in variation of the alligator population shown in Figure 1. Use specific values in your answer. [3 marks]
- (ii) Using the information presented in Figure 1, determine the carrying capacity of the Black River. [1 mark]
- (iii) In which year was the carrying capacity of the Black River MOST likely achieved? [1 mark]
- (c) A teacher was overheard saying to her class that "... all living organisms need a source of phosphorous. ... However there is one major difference between the phosphorous cycle and the carbon cycle that can have a profound effect on the amount of available phosphorous and the productivity of an ecosystem. We need to be careful in what we do because humans can have a major impact on the phosphorous cycle".

Suggest FOUR reasons to support the statement made by the teacher.

[12 marks]

Total 20 marks

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2. (a) The results of a sampling procedure to determine the species of mangroves in three ecosystems, A, B and C, are presented in Table 1.

TABLE 1: RESULTS OF POPULATION SAMPLING

Species of Mangroves	Number of Mangrove Trees in Ecosystem		
	A	B	C
Red mangrove	50	40	12
Black mangrove	38	6	8
White mangrove	12	18	22
Buttonwood mangrove	60	70	40
Species diversity	3.379		3.005

- (i) The species diversity for EACH of Ecosystems A and C is indicated in Table 1. Calculate the species diversity for Ecosystem B using the equation below.

$$D = \frac{N(N-1)}{\sum n(n-1)}$$

[5 marks]

- (ii) Rank the three ecosystems in DECREASING order of species diversity.

[1 mark]

- (iii) Which of three ecosystems is the MOST stable?

[1 mark]

- (iv) Suggest FOUR reasons to support your answer to (iii) above.

[4 marks]

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- (b) Table 2 shows the results of an ecological study where three amphibians were observed in a grassland ecosystem. Study the information in Table 2 and answer the question that follows.

TABLE 2: RESULTS OF ECOLOGICAL STUDY

Parameter Being Observed	Amphibian A	Amphibian B	Amphibian C
Food type	Ants and flies	Termites	Ants and flies
Habitat	Moist grassy areas	Moist grassy areas	Moist grassy areas
Feeding behaviour	Daytime feeder	Night-time feeder	Daytime feeder

Suggest THREE inferences that can be made about the competition taking place among the amphibians in the grassland ecosystem. [9 marks]

Total 20 marks

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3. (a) Define the term 'ecological niche'. [3 marks]

(b) Figure 2 illustrates a predator-prey relationship in a research plot in a savannah. Study Figure 2 and answer the questions below.

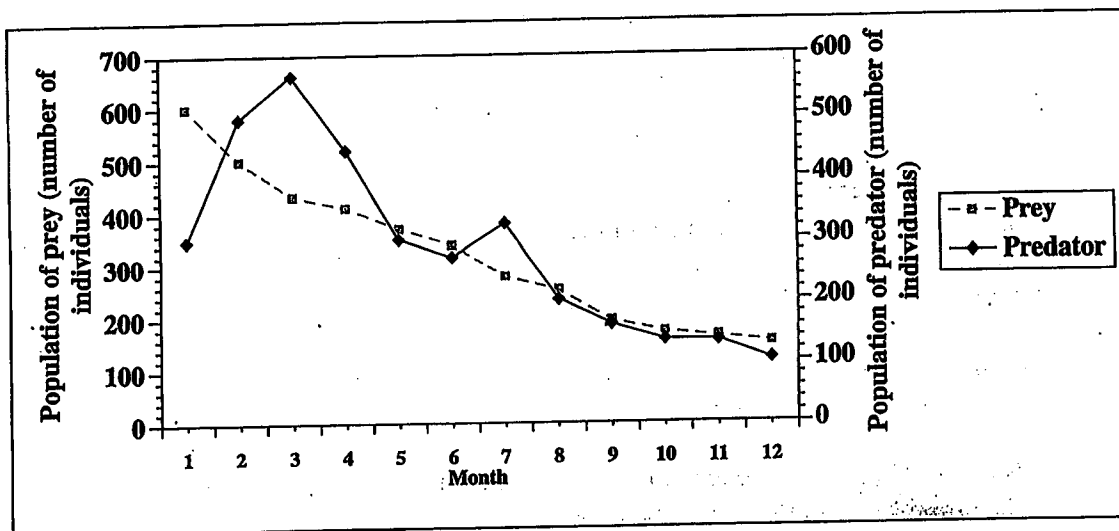


Figure 2. Predator-prey relationship in a research plot in a savannah

(i) Describe the trend shown in Figure 2. Include specific values in your answer. [3 marks]

(ii) In which month was size of the predator population equal to 200? [1 mark]

(iii) What was the size of the prey population in Month 11? [1 mark]

(iv) Suggest TWO reasons why the predator population is successfully regulating the population of the prey. [6 marks]

(c) "Every living organism has limits to the environmental conditions it can endure."

Provide THREE reasons to support the statement above.

[6 marks]

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. (a) Define the term 'poverty'. [2 marks]
- (b) Study Figure 3 which shows the percentage of the world's population that is affected by lack of access to selected social facilities. Given that the world's total population is 6.7 billion, answer the questions that follow.

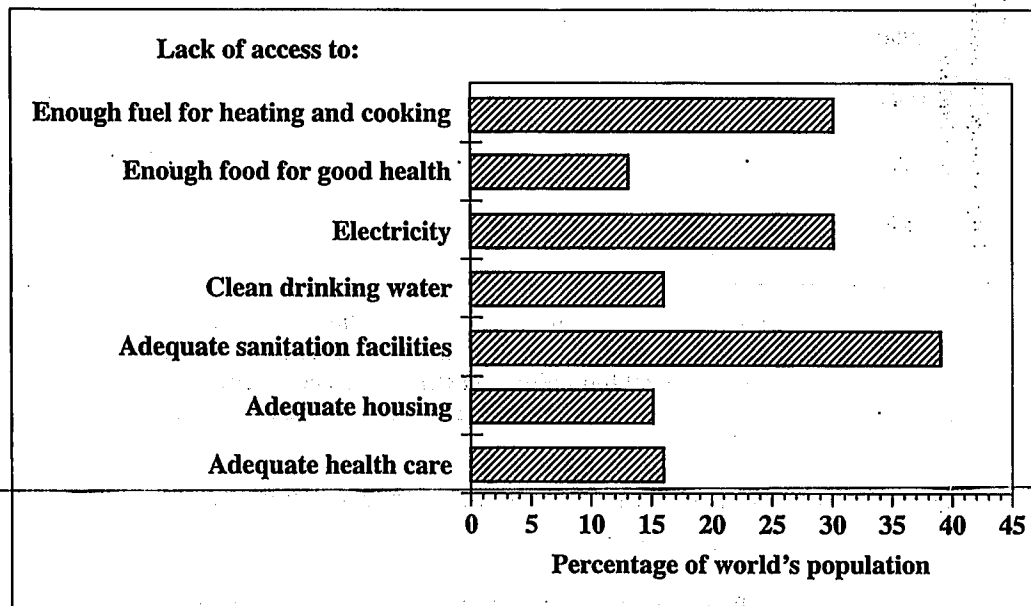


Figure 3. Percentage of the world's population that is affected by lack of access to selected social facilities

- (i) What percentage of the world's population lacks access to adequate sanitation facilities? [1 mark]
- (ii) Calculate the number of people who lack access to adequate housing and electricity. [2 marks]
- (iii) How many more people lack access to clean drinking water than to enough food for good health? [3 marks]
- (c) At an environmental seminar, you overhear two statements in a conversation.
- Statement 1: "Caring about the environment is often a challenge for poor people."
Statement 2: "Poverty has harmful environmental and health effects."
- (i) Suggest ONE reason why someone might say Statement 1. [4 marks]
- (ii) Provide FOUR reasons to support Statement 2. [8 marks]

Total 20 marks

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5. (a) The 'ecological footprint' of a country is the amount of biologically productive land and water needed to supply the people in an area with renewable resources and to absorb and recycle the wastes and pollution caused by resource use. Study Figure 4 which shows total and per capita ecological footprint for selected countries and answer the questions that follow.

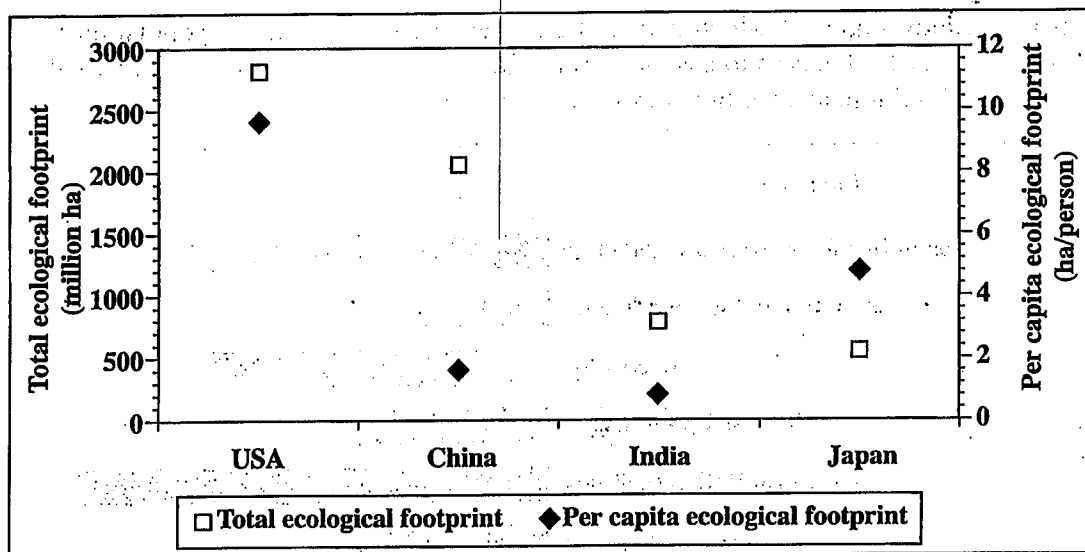


Figure 4. Total and per capita ecological footprint for selected countries

- (i) What is the **total** ecological footprint for the country with the **LARGEST** per capita footprint? [1 mark]
 - (ii) What is the per capita ecological footprint for the country with the **SMALLEST** total footprint? [1 mark]
 - (iii) Describe the trends observed in Figure 4, using specific values in your answer. [4 marks]
 - (iv) China has a relatively small per capita ecological footprint, yet has the second highest total ecological footprint. Suggest **ONE** reason for this apparent discrepancy. [2 marks]
- (b)
- (i) Describe **TWO** ways in which people impact negatively on the environment, using **THREE** specific examples to illustrate **EACH** way. [6 marks]
 - (ii) For **EACH** way in which people impact negatively on the environment as described in (b) (i) above, suggest **ONE** way in which this negative result may be mitigated. Relevant, specific examples are required in your response. [6 marks]

Total 20 marks

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6. (a) An insert is provided for this question.

On the insert provided, construct an age structure diagram from the data presented in Table 3. In your diagram, use a horizontal scale of 2 cm to represent 2 million people.

[5 marks]

TABLE 3: AGE GROUP DATA FOR THE POPULATION OF COUNTRY X

Age Group	Number of People (millions)	
	Female	Male
0 – 14	15	12
15 – 30	7.4	5
31 – 50	5.6	3.8
51 – 70	2.6	2.4
71 and over	1.4	1.2

- (b) (i) Suggest a country that would have an age structure diagram similar to the one developed in (a) above. [1 mark]

- (ii) Would the estimate of the likely rate of population growth in Country X be high, medium, low or negative? [1 mark]

- (iii) With reference to the age structure diagram developed in Part (a), explain your response to (b) (ii). [3 marks]

- (c) (i) Define the term 'total fertility rate'. [2 marks]

- (ii) Study Table 4 and answer the questions below.

TABLE 4: AVERAGE GLOBAL TOTAL FERTILITY RATE (CHILDREN/WOMAN)

	Developed Countries	Developing Countries
1950	2.5	6.5
2007	1.6	2.9

- a) What is the trend over time in average global total fertility rate shown in Table 4? [2 marks]
- b) Discuss THREE factors that would have contributed to the trend observed in Table 4. [6 marks]

Total 20 marks

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MODULE 3

Answer ANY TWO questions.

7. (a) Figure 5 represents a traditional agricultural practice of indigenous peoples in a Caribbean country.



Figure 5. Traditional agricultural practice

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- (i) Identify the traditional agricultural practice in Figure 5. [1 mark]
- (ii) Outline how this type of agriculture practice is carried out. [4 marks]
- (iii) "Indigenous forest dwellers often cause tropical forest destruction because of their traditional agricultural practice."

Do you agree with this statement? Provide THREE reasons to support your answer. [6 marks]

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8. (a) Figure 7 shows the amount of fish harvested from traditional marine fishing grounds in Country X. Study Figure 7 and answer the questions that follow.

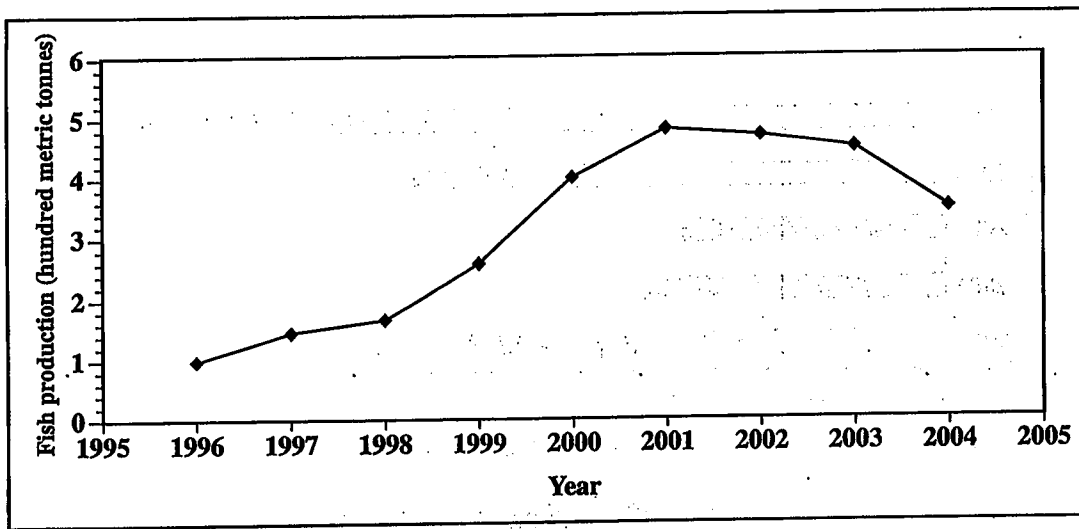


Figure 7. Fish harvest for Country X from traditional fishing grounds

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- (i) Describe the trends in fish harvest shown in Figure 7. [3 marks]
- (ii) In what year was the maximum yield of fish obtained? [1 mark]
- (iii) Suggest TWO reasons for the trends shown in Figure 7. [4 marks]
- (iv) At what production level would you recommend that the country continue to harvest its fish resource? Provide THREE reasons for your recommendation. [6 marks]
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- (b) Figure 6 shows the percentage employment in a Caribbean country in two natural resource sectors: fisheries and forestry. Study Figure 6 and answer the questions that follow.

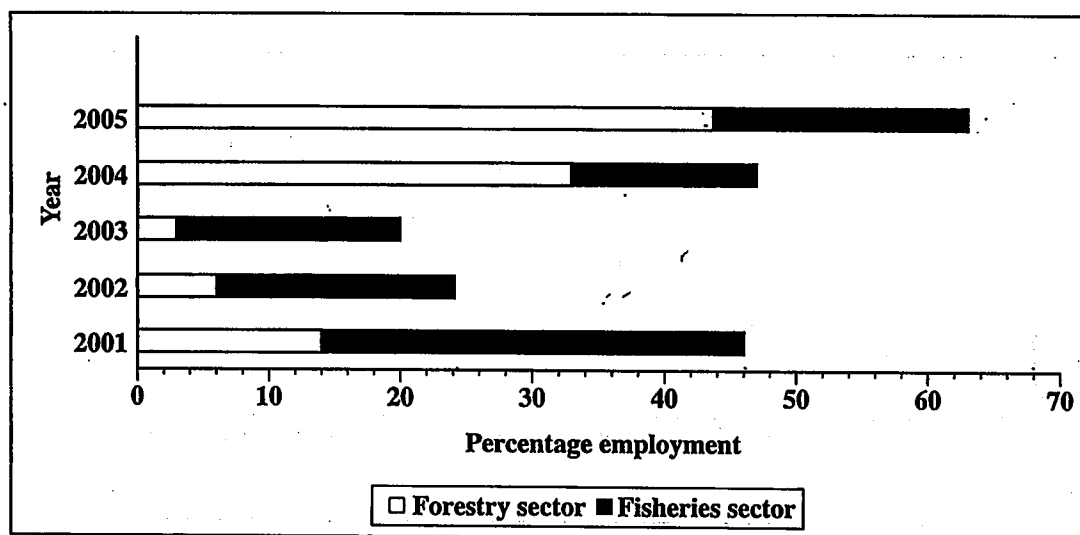


Figure 6. Percentage employment in the fisheries and forestry sector in a Caribbean country

- (i) Describe the trend illustrated in Figure 6. [3 marks]
- (ii) In which year is the employment in the fisheries sector GREATEST? [1 mark]
- (iii) What is the TOTAL percentage employment of the forestry sector in 2003 and 2004? [1 mark]
-
- (c) A logging company applied to the government of the Caribbean country, R, for permission to harvest logs from a part of the forest. They were advised by the Environmental Protection Agency (EPA) that an Environmental Impact Assessment (EIA) was needed before their request could be considered since the conservation of the country's forests was a priority.
- Justify the request of the EPA for the Environmental Impact Assessment. Give TWO points in your answer. [4 marks]

Total 20 marks

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- (b) Figure 8 shows a remote location in a Caribbean country where work has begun for the exploitation of a gold reserve. The location of the gold reserve is a heavily forested and hilly area with few population centres.

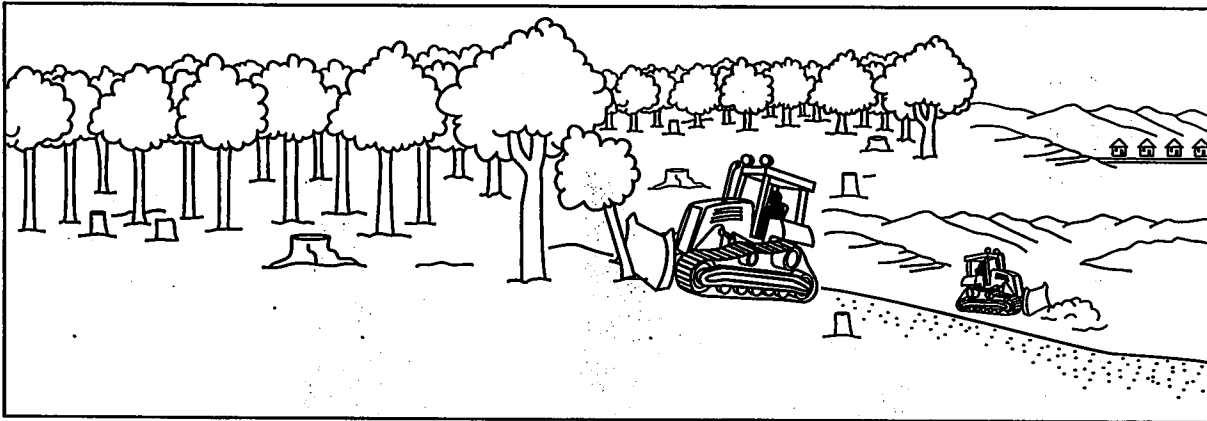


Figure 8. Location of gold reserve in a Caribbean country

Outline TWO factors that the government of the Caribbean country may have considered before making the decision to grant permission to private investors to exploit the gold resource. [6 marks]

Total 20 marks

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9. (a) Figure 9 shows the contributions to the economy made by natural resources over a three-year period for a Caribbean country, P. Study Figure 9 and answer the questions that follow.

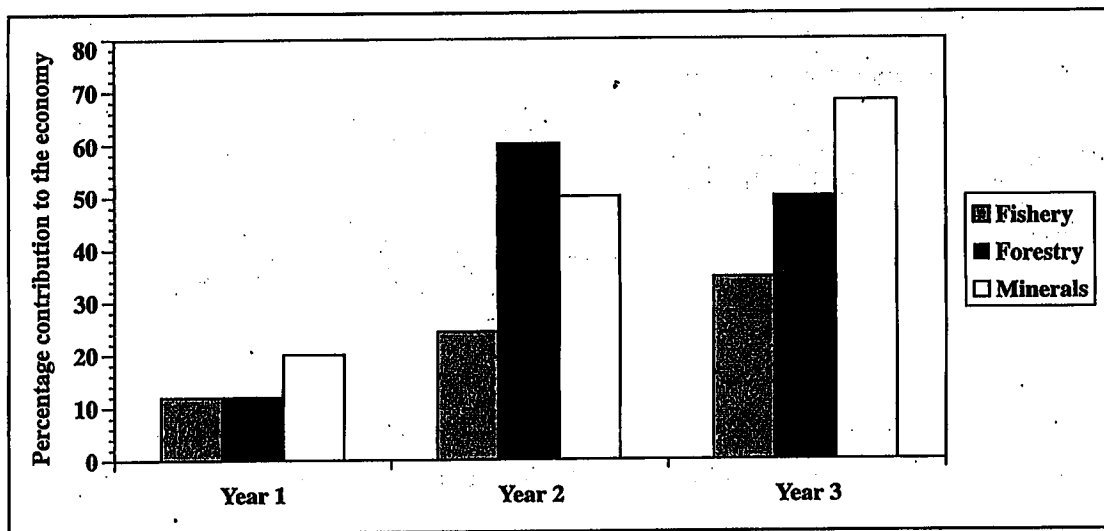


Figure 9. Contributions from natural resources

- (i) Describe the trend in the contribution made by natural resources in Country P. [3 marks]
- (ii) What was the percentage contribution made by forestry in Year 1? [1 mark]
- (iii) What was the difference in percentage contributions made by minerals in Year 1 and Year 3? [1 mark]
- (b) In the Caribbean country, P, which has many beaches, many people do not see the need for conservation and management of beaches as a natural resource. Suggest THREE reasons to convince a group of students that the conservation of beaches as a natural resource is necessary for the country. [9 marks]
- (c) Justify the usefulness of the following in the conservation and management of beaches in the Caribbean:
- (i) Education and awareness [3 marks]
- (ii) Legislation [3 marks]

Total 20 marks

END OF TEST

TEST CODE **02120020**
MAY/JUNE 2010

ENVIRONMENTAL SCIENCE

PAPER 02

Centre Number

Candidate's Number

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